

Roll No.

TCS-201

**B. TECH. (SECOND SEMESTER)
END SEMESTER EXAMINATION, June, 2023**

PROGRAMMING FOR PROBLEM SOLVING

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question (except question no. 2).

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Draw a flowchart to input an alphanumeric string. Find and print sum of all digits of that string.

Inputted sting : My mobile number is 345 and landline number is 262.

Output : 22. (CO1, CO2, CO3)

- (b) Write a program to input a multiword string and print length of each word.

Inputted sting : Dehradun is the capital of Uttarakhand.

Output : 8 2 3 7 2 11. (CO1, CO2, CO3)

- (c) Write a program to input a string and delete an inputted character from whole string. Print the final string. (Using pointers)

Sample Input :

Inputted sting : Dehradun is the capital of Uttarakhand.

Inputted character : a

Output string : Dehrdun is the cpitl of Uttrkhnd.

P. T. O.

2. Find output of the following codes : (CO1, CO2, CO3, CO4, CO5, CO6)

(1) #include<stdio.h>

```
void main()
{
    int i = 10;
    void *p =&i;
    printf("%d", (int)*p);
}
```

(2) #include<stdio.h>

```
void main()
{
    int*p,a=10;
    p=&a;
    *p+=1;
    printf("%d %d", *p, a);
}
```

(3) #include<stdio.h>

```
void main()
{
    char str1[20] = "Hello";
    char str2 [20] = "World";
    printf("%s\n", strcpy(str2,
    strcat(str1, str2)));
}
```

- (4) What will be the output of the following Python code ?

```
number = 5
while number <= 5:
    if number < 5:
        number = number + 1
print(number)
```

- (5) #include <stdio.h>

```
char *myfunc(char*p)
{
    p+=3;
    return(p);
}
void main()
{
    char *x, *y;
    x = "AYURVEDA";
    y = myfunc(x);
    printf("%s", y);
}
```

- (6) #include <stdio.h>

```
void test(int *a, int *b)
{
    a = b;
    *a = 15;
}
int x = 10, y = 20;
int main()
{
    test(&x, &y);
    printf("%d %d" x, y);
    return 0;
}
```

(7) #include <stdio.h>

```
struct employee
```

```
{
```

```
    char *empname;
```

```
    int salary;
```

```
};
```

```
int main()
```

```
{
```

```
    struct employee e, e1;
```

```
    e.empname = "Sridhar";
```

```
    e1 = e;
```

```
    printf("%s, %s", e.empname,  
    e1.empname);
```

```
    return 0;
```

```
}
```

(8) What is the output of this program in the text file?

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    FILE *pFile;
```

```
    char c;
```

```
    pFile = fopen("sample.txt", "w+");
```

```
    for (c = 'A'; c <= 'E'; c++)
```

```
    {
```

```
        putc(c, pFile);
```

```
    }
```

```
    fclose(pFile);
```

```
    return 0;
```

```
}
```

```
(9) #include<stdio.h>
int main()
{
    struct forest
    {
        int trees;
        int animals;
    }F1, * F2;
    F1.trees=1000;
    F1.animals=20;
    F2=&F1;
    printf("%d", F2.animals);
    return 0;
}
```

(10) What will be the last value of x in the following Python code ?

```
x = 0
for i in range(10):
    for j in range(-1, -10, -1):
        x += 1
    print(x)
```

3. (a) Write short notes on the following :

(CO4)

- (i) Structure vs. Array
- (ii) Bitfield
- (iii) Structure within structure
- (iv) Self-referential structure
- (v) Define Union

P. T. O.

(b) Create a structure Customer with the data fields customer_name, account_number and account_balance of customer. Store information of n customers : (CO4)

(i) Write a function to print name of all the customers having balance less than ₹ 5,000.

(ii) Write a function to add ₹ 1,000 in the account_balance of all the customers having balance more than ₹ 10,000 and print the updated details of all the customers.

(c) Create a structure Employee with the data fields employee_name, employee_id, employee_salary and employee_year of experience. Input details of n employees and arrange the employees in descending order of their year of experience. Finally print detail of all the employees.

(CO4)

4. (a) Explain the following : (CO5)

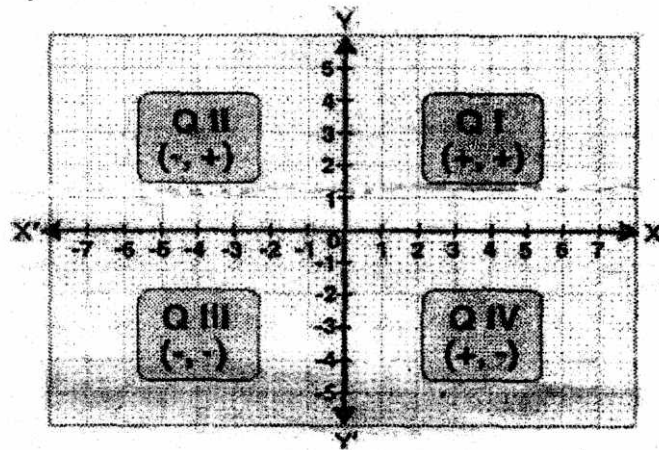
(i) Five library functions of File Handling

(ii) Random file accessing methods

(b) Write a program to store details of five students into a file (DATA.txt). Detail of student include student_name, student_rno, student_section and marks in three subjects. Read this file and display the content of file along with average marks. (CO5)

(c) Write a program to store n integer numbers into a file (INTEGER.txt). Read this file and write reverse of each number into another file (REVERSE.txt). Display content of both the files. (CO5)

5. (a) Describe applications and features of Python. Explain the operators in Python. (CO6)
- (b) Write a Python code to accept a coordinate of a point in a XY-coordinate system and determine in which quadrant the point lies : (CO6)



- (c) Write a Python code to find sum of the following series : (CO6)
- $$1 + 2*1 + 3*2*1 + 4*3*2*1 + \dots + n*(n-1)*(n-2)\dots*1$$

Roll No.

TMA-201

B. TECH. (SECOND SEMESTER)
END SEMESTER EXAMINATION, June, 2023
ENGINEERING MATHEMATICS—II

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Obtain the general solution of the differential equation : (CO1)

$$(D^2 + 2)y = e^x \cos x + x^2 e^{2x}$$

(b) Solve the homogeneous differential equation : (CO1)

$$x^2 \frac{d^2 y}{dx^2} - x \frac{dy}{dx} - 3y = x^2 \log x$$

(c) Use method of variation of parameter to solve : (CO1)

$$\frac{d^2 y}{dx^2} - 2 \frac{dy}{dx} = e^x \sin x$$

2. (a) Find the Laplace transform of : (CO2)

$$\frac{1 - \cos t}{t^2}$$

P. T. O.

(b) Find the Laplace transform of the following function of period $2a$:

(CO2)

$$f(t) = \begin{cases} \frac{h}{a}t & , 0 < t < a \\ \frac{h}{a}(2a - t), & a < t < 2a \end{cases}$$

(c) State Convolution Theorem and use it to evaluate :

(CO2)

$$L^{-1} \left[\frac{s}{(s^2 + a^2)^2} \right]$$

3. (a) Find the Fourier series for the function :

(CO3)

$$f(x) = \frac{(\pi - x)^2}{4}; \quad 0 < x < 2\pi$$

(b) Find the Fourier expansion for the function $f(x) = x - x^2$ in the interval $(-\pi, \pi)$. Hence, deduce that :

(CO3)

$$\frac{1}{1^2} - \frac{1}{2^2} + \frac{1}{3^2} - \frac{1}{4^2} + \dots = \frac{\pi^2}{12}$$

(c) Find the Fourier cosine series expansion of the periodic function defined by :

(CO)

$$f(t) = \sin\left(\frac{\pi t}{l}\right), \quad 0 < t < l$$

4. (a) Find the general solution of the following PDE :

(CO4)

$$(D^2 - 3DD' + 2D'^2)z = e^{2x-y} + e^{x+y} + \cos(x + 2y);$$

$$D \equiv \frac{\partial}{\partial x} \text{ and } D' \equiv \frac{\partial}{\partial y}$$

(3)

- (b) Solve the one-dimensional wave equation $\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$, with boundary conditions :

(CO4)

$$u(0, t) = 0$$

$$u(l, t) = 0$$

$$u(x, 0) = f(x)$$

and

$$\left(\frac{\partial u}{\partial t} \right)_{t=0} = g(x).$$

- (c) Solve the following PDE using the method of separation of variables :

(CO4)

$$\frac{\partial^2 u}{\partial x \partial t} = e^{-t} \cos(x),$$

where $u(x, 0) = 0$ and $u_t(0, t) = 0$.

5. (a) Prove that :

(CO5)

$$\int_{-1}^1 [P_n(x)]^2 dx = \frac{2}{2n+1}$$

- (b) Prove the following recurrence relation of Bessel's equation :

(CO5)

$$\frac{d}{dx} [x^n J_n(x)] = x^n J_{n-1}(x)$$

- (c) Prove that :

(CO5)

$$J_{\frac{1}{2}}(x) = \sqrt{\frac{2}{\pi x}} \sin x \text{ and } J_{-\frac{1}{2}}(x) = \sqrt{\frac{2}{\pi x}} \cos x$$

Roll No.

THU-201

B. TECH. (SECOND SEMESTER)
END SEMESTER EXAMINATION, June, 2023
ADVANCED PROFESSIONAL COMMUNICATION

Time : Three Hours

Maximum Marks : 100

- Note :** (i) All questions are compulsory.
(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.
(iii) Total marks in each main question are **twenty**.
(iv) Each sub-question carries 10 marks.

1. (a) Explain in detail the different methods of delivering a presentation. (CO5)
- (b) Differentiate between a Resume and CV. What are the components of a Resume ? (CO5)
- (c) Differentiate between the following : (CO5)
- (i) JAM and Extempore Speech
- (ii) Impromptu Delivery and Extemporaneous Delivery

P. T. O.

2. (a) Below mentioned are the components of the Resume prepared by Mr. Samuel : (CO3, CO5)

- (i) CAREER OBJECTIVE
- (ii) EDUCATIONAL QUALIFICATION
- (iii) CORE COMPETENCIES
- (iv) CERTIFICATIONS
- (v) TECHNICAL EXPERTISE
- (vi) PROFESSIONAL EXPERIENCE
- (vii) TRAININGS/WORKSHOPS
- (viii) INTERESTS

(1) What do you mean by Core Competencies ? Mention your core competencies.

(2) Frame a career objective for Mr. Samuel as he is preparing his resume for the job of a Web Developer. (Take assumptions and add necessary details)

(b) Briefly describe the process of Group Discussion. Also list the type of GD topics with examples. (CO3, CO5)

(c) What do you mean by "An art of conversation in formal environment" ? (CO3, CO5)

3. (a) What is a Technical Report ? Briefly describe the *three* types of technical reports. (CO4)

(b) What do you mean by Research Methodology ? What are the objectives of Research ? (CO4)

(c) What is a Technical Proposal ? What are the components of a Technical Proposal ? (CO4)

4. (a) What is Précis Writing ? State key features of Précis Writing.

(CO1, CO2)

(b) Write Descriptive paragraph in about 100-150 words on "An Engineer".

(CO1, CO2)

(c) Use the following Phrasal verbs in sentences to clarify the meaning :

(CO1, CO2)

(i) Set out

(ii) Look up

(iii) Look over

(iv) Call off

(v) Catch on

5. (a) Provide :

(CO1, CO2)

(i) Oxymoron meaning "Only option left"

(ii) Phrasal Verb meaning "To start doing a particular activity"

(iii) Idiom meaning "Honest"

(iv) Oxymoron meaning "Normal but slightly different"

(v) Phrasal Verb meaning "Rely on"

(b) Use the following idioms in sentences to clarify the meaning :

(CO1, CO2)

(i) A White Elephant

(ii) All at Sea

(iii) To sell like hot cakes

(iv) A close shave

(v) Break a leg

P. T. O.

(c) Answer the following questions : (True/False) (CO1, CO2)

- (i) While starting to write the body of the letter, full name of the recipient needs to be written.
- (ii) The correct way of signing a letter off is Best Regards/Best regards.
- (iii) The signatures are to be put above/under the name of the sender.
- (iv) Writing references in any document, makes the document-in nature/ (Which C of the 7 C's ?)
- (v) In a Just a minute session, the speaker is always/never given time to prepare or organize the thoughts.

Roll No.

TEC-201

**B. Tech. (SECOND SEMESTER)
END SEMESTER EXAMINATION, June, 2023**

(All Branches)

BASIC ELECTRONICS ENGINEERING

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Simplify the given Boolean expression using Boolean rules : (CO1)

(i) $F = ABD + CD + ACD + ABC + ABCD$

(ii) $F = (A + C)(AD + AC + C)$

(b) Obtain the minimal SOP expression for the following function and implement the same using universal gate : (CO1)

(i) $F(A, B, C, D) = \sum m(0, 2, 3, 5, 7, 8, 13) + \sum d(1, 6, 12)$

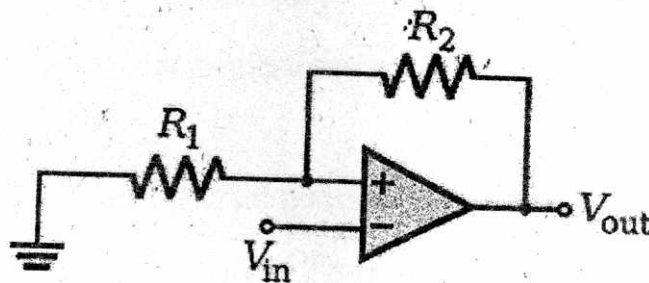
(ii) $F(A, B, C, D) = \sum m(1, 4, 7, 8, 9, 11) + \sum d(0, 3, 5)$

P. T. O.

- (c) What do you understand by basic and universal gates ? List all the basic and universal gates with their symbol and truth table. Realize EX-NOR gate using NAND gates only. (CO1)
2. (a) (i) Explain the doping and types of doping in semiconductors.
- (ii) The intrinsic carrier concentration for silicon is $1.5 \times 10^{10}/\text{cm}^3$. If mobility of electrons and holes are $1300 \text{ cm}^2/\text{V-s}$ and $450 \text{ cm}^2/\text{V-s}$ respectively. What is the conductivity of silicon (intrinsic) ? If silicon is doped with 10^{18} boron atoms per cm^3 , what will be its conductivity ? (CO2)
- (b) (i) Distinguish between Avalanche breakdown and Zener breakdown.
- (ii) A Si diode has reverse saturation current of $2.5 \mu\text{A}$ at 300K . Find forward voltage to achieve a forward current of 10 mA .
($V_T = 26 \text{ mV}$) (CO2)
- (c) Explain the forward and reverse bias condition of a P-N junction diode with V-I characteristics. (CO2)
3. (a) Explain the working of a bridge rectifier with a suitable circuit diagram and waveforms. Also, derive expressions for its RMS and average currents, efficiency and ripple factor. (CO3)
- (b) A 220 V , 60 Hz voltage is applied to a center tapped step-down transformer of $22 : 1$ with a load of $k\Omega$ connected across the output of two-diode full-wave rectifier. Assume diodes to be ideal and determine the (i) peak, r.m.s. and d.c. currents, (ii) d.c. power delivered to the load, (iii) input a.c. power, (iv) efficiency of the rectifier, (v) a.c. ripple voltage across the load and its frequency. (CO3)

(3)

- (c) With the help of circuit diagrams and waveforms, explain the working of capacitor filter in case of a half wave rectifier circuit. (CO3)
4. (a) Explain about input and output characteristics of common Emitter Configuration of transistor. (CO4)
- (b) Explain about input and output characteristics of Common Base Configuration of transistor. (CO4)
- (c) (i) Derive the relationship between α , β and γ .
- (ii) Calculate α and β of a transistor for which I_C is 5 mA, I_B is 50 μ A and I_{CBO} is 1 μ A. (CO4)
5. (a) (i) Explain the concept of virtual ground.
- (ii) Determine the value of feedback resistor R_2 , such that the close loop gain is 10 and $R_1 = 5 \text{ k}\Omega$, in the circuit given below : (CO6)



- (b) (i) What is an operational amplifier ? Draw the block diagram of op-amp.
- (ii) Draw and explain the input/output characteristics of an ideal op-amp. (CO5, CO6)
- (c) With suitable circuit diagrams, derive an expression for output voltage of summing amplifier. (CO5, CO6)

Roll No.

BTE-201

**B. TECH. (BIOTECH) (SECOND SEMESTER)
END SEMESTER EXAMINATION, June, 2023**

ADVANCED ORGANIC CHEMISTRY

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Elaborate the distillation method with suitable diagram. (CO1)
- (b) Define the term chromatography and give the classification of chromatography. (CO1)
- (c) Explain the solvent extraction technique with suitable diagram. (CO1)
2. (a) Analyze the following terms with suitable example : (CO2)
 - (i) Resonance
 - (ii) Hyperconjugation
- (b) Discuss the orientation effect in aromatic substitution reactions. (CO2)
- (c) Elaborate the nitration of benzene with suitable example. (CO2)

P. T. O.

(2)

3. (a) What do you mean by green chemistry and elaborate its twelve principles? (CO3)
- (b) Explain the term nanoparticles and give the classification and application of nanoparticles. (CO3)
- (c) Discuss the various approaches involved in the synthesis of nanoparticles. Explain the various adverse effects associated with the use of chemicals. (CO3)
4. (a) Discuss the principle working technique and application of column chromatography. (CO4)
- (b) Explain the principle and working technique of paper chromatography. (CO4)
- (c) Analyze the principle application and working technique of thin layer chromatography. (CO4)
5. (a) What do you mean by the term carbohydrate and give its classification? (CO5)
- (b) Explain the physical and chemical properties of glucose. (CO5)
- (c) What do you mean by Killiani Fischer synthesis and ruff degradation reaction in carbohydrate chemistry? (CO5)

Roll No.

TCH-201

**B. TECH. (SECOND SEMESTER)
END SEMESTER EXAMINATION, June, 2023**

(All Branches)

ENGINEERING CHEMISTRY

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) What are bonding and antibonding molecular orbitals ? Explain the condition for the formation of molecular orbitals from atomic orbitals with the help of example. (CO1)
- (b) Explain different vibrational modes in IR spectroscopy. Write applications of IR spectroscopy. (CO1)
- (c) Define bond order and its significance. In case of N_2 and N_2^+ which has higher bond dissociation energy and why ? (CO1)

P. T. O.

2. (a) Describe the lime soda process for water softening. Give the chemistry involved during the softening. Why does magnesium bicarbonate require double amount of lime for softening? (CO2)
- (b) (i) Differentiate between Scales and Sludges. (CO2)
- (ii) A water sample contains 408 mg of CaSO_4 per litre. Calculate the hardness in terms of CaCO_3 equivalents.
- (c) A zeolite bed exhausted after completely removing the total hardness of 10,000 litre of hard water. The zeolite was regenerated using 8.0 litre of NaCl containing 150 g/l of NaCl. Calculate the hardness of water. (CO2)
3. (a) What do you mean by conducting polymers? Discuss classification of conducting polymer. (CO3)
- (b) Classify polymers on the basis of tactility and intermolecular forces. (CO3)
- (c) Explain the terms polymerization, functionality and degree of polymerization (DP). Calculate DP of PVA sample having 100000 average molecular weight. (CO3)
4. (a) What are biofuels? Write a note on process for production of biogas and factors affecting it. (CO4)
- (b) Define calorific value. Calculate GCV and NCV of a fuel sample from following data : (CO4)
- Mass of fuel = 0.85 gm
- Water equivalent mass of calorimeter = 2500 gm

(3)

Amount of water taken in calorimeter = 650 gm

Rise in temperature = 2.5°C

Fuse wire correction = 10 calories

Cooling correction = 0.003°C

Acid correction = 50 cal

% of H = 8

Given latent heat of condensation of steam is 580 cal/gm.

- (c) (i) Write properties of good fuels. (CO4)
(ii) Explain the classification of fuels with example.

5. (a) What is electrode potential ? Discuss Danial cell by giving suitable diagram. (CO5)

(b) What is electrochemical series ? Discuss applications of electrochemical series. (CO5)

(c) For the cell : (CO5)



$$[E_{\text{Mg}^{2+}/\text{Mg}}^{\circ} = -2.37 \text{ V and } E_{\text{Cu}^{2+}/\text{Cu}}^{\circ} = 0.34]$$

- (i) Identify the cathode and the anode.
(ii) Write the oxidation and reduction reactions.
(iii) Calculate the EMF of cell.

Roll No.

TEV-201

**B. TECH. (SECOND SEMESTER)
END SEMESTER EXAMINATION, June, 2023**

ENVIRONMENTAL SCIENCE

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Define Environmental Science. Discuss its multidisciplinary nature.
(CO1 & CO3)
- (b) Define Ecosystem. Explain its components. (CO1 & CO3)
- (c) What is ecological pyramid ? Discuss its various types with example.
(CO1 & CO3)
2. (a) Define Natural Resource. Discuss any *one* natural resource in detail.
(CO3 & CO6)
- (b) What is rainwater harvesting ? Discuss various components of rooftop rainwater harvesting structure. (CO3 & CO6)

P. T. O.

(2)

- (c) Explain the environmental problems associated with modern agricultural techniques. How recent advancement in agriculture sector will help to overcome the problem ? (CO3 & CO6)
3. (a) Define solid waste. Discuss various solid waste disposal methods. (CO2 & CO4)
- (b) Define soil pollution and discuss the sources and effect of soil pollution. (CO2 & CO4)
- (c) "The industrial developments have caused positive as well as negative effects on environment". Comment. How can public awareness improve the environmental degradation ? (CO2 & CO4)
4. (a) What is Ecological Succession ? Discuss all steps involved in it. (CO4 & CO5)
- (b) Write short notes on the following : (CO4 & CO5)
- (i) Conservation of Biodiversity
- (ii) Use of forest
- (c) Define pollution. Describe effects and control of air pollution. (CO4 & CO5)
5. (a) Define disaster. Explain any *one* recent disaster occurred in Uttarakhand in detail. (CO1 & CO3)
- (b) Explain greenhouse effect. What are the consequence of greenhouse effect ? (CO1 & CO3)
- (c) Write short notes on the following : (CO1 & CO3)
- (i) Earthquake
- (ii) Acid rain

Roll No.

TEV-222

B. C. A./B. SC. (IT)/B. SC. (CS)

(SECOND SEMESTER)

END SEMESTER EXAMINATION, June, 2023

ENVIRONMENTAL STUDIES

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) What are ecological pyramids ? Explain with types of pyramids and examples. (CO1)
- (b) What are decomposers ? Discuss the role of decomposers in an ecosystem. (CO1)
- (c) Define ecosystem. How does energy and nutrient flow in an ecosystem ? (CO1)
2. (a) What is Deforestation ? Discuss various effects of deforestation. (CO4)

P. T. O.

(2)

- (b) Discuss the role of an individual in the conservation of natural resources. (CO4)
- (c) Define Biodiversity. Explain *In-situ* and *Ex-situ* conservation of biodiversity. (CO4)
3. (a) Define air pollution. Discuss sources and effects of air pollution. (CO2, CO5)
- (b) Describe the processes adopted for solid waste management. (CO2, CO5)
- (c) What is water pollution ? What are the effects of water pollution ? (CO2, CO5)
4. (a) Define disaster. Discuss various methods to reduce the impact of flood in detail. (CO3, CO4)
- (b) Write short notes on the following : (CO3, CO4)
- (i) Global warming
- (ii) Acid rain
- (c) What is sustainable development ? Discuss necessary conditions required to achieve sustainable development. (CO3, CO4)
5. (a) Differentiate between the following : (CO1, CO3)
- (i) Primary and Secondary air pollutants
- (ii) London and Los Angeles smog
- (b) Define food chain and food web. Discuss its significance in a ecosystem. (CO1, CO3)
- (c) Discuss the advantages and disadvantages of dams. (CO1, CO3)

Roll No.

BTS-201

**B. TECH. (BIOTECHNOLOGY) (SECOND SEMESTER)
END SEMESTER EXAMINATION, June, 2023**

ENTREPRENEURIAL BIOTECHNOLOGY

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Elaborate on the concerns related to bio-safety with reference to research in biotechnology. (CO1)
(b) What are the different biosafety levels ? Elaborate in detail. (CO1)
(c) Elaborate on the environmental issues related to bioethics. (CO1)
2. (a) Discuss some of the important types of bio-business in India. (CO2)
(b) Explain Entrepreneurship. What are the different types of Entrepreneurship ? (CO2)
(c) What is SWOT analysis ? Elaborate in detail. (CO2)
3. (a) Explain in detail the different types of Hydroponics systems. (CO3)
(b) Elaborate in detail on the technique of plant tissue culture. (CO3)
(c) What are nutraceuticals ? Explain their role in health and nutrition. (CO3)

P. T. O.

(2)

4. (a) What is biocontrol ? Discuss the different advantages it has over synthetic fertilizers. (CO4)
- (b) Elaborate on organic farming and discuss its advantages. (CO4)
- (c) What are fermented foods ? Discuss the various health benefits of probiotic food. (CO4)
5. (a) Discuss some of the challenges faced by the Biotechnology Business in India. (CO5)
- (b) Discuss the role of Biotech Parks in encouraging bio-business setup. (CO5)
- (c) Elaborate on the role of the Department of Biotechnology in promoting bio-business startup. (CO5)

Roll No.

TPH-201

**B. TECH. (SECOND SEMESTER)
END SEMESTER EXAMINATION, June, 2023**

(All Branches)

ENGINEERING PHYSICS

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Explain reason of formation of circular rings in Newton's ring experiment, deduce the expression of diameter of dark ring and find the ratio of diameter of dark rings in air and in the transparent liquid of refractive index (μ). (CO1)
- (b) In Fraunhofer diffraction due to single slit derive the expression for resultant intensity and also write the ratio of intensities for central maxima and secondary maximas. (CO1)
- (c) Derive the expression for the fringe width in Fresnel Biprism experiment and write the condition to obtain sustain and distinct interference pattern. (CO1)

P. T. O.

2. (a) Explain the working principle of laser and discuss the construction and working of Ruby laser. (CO2)
- (b) Explain Law of Malus. Derive the necessary expression to discuss that circularly polarized light is special case of elliptically polarized light. (CO2)
- (c) (i) A pulsed laser is constructed with ruby crystal containing a total of 3×10^{19} chromium ions. If laser emits $6940 \text{ \AA}$ find the energy of one photon and energy of pulse (assuming all the number of chromium ions participate in emission of a pulse). (CO2)
- (ii) Calculate the specific rotation if the plane of polarization is turned through 30° travelling 20 cm length of 20% sugar solution.
3. (a) Discuss Michelson-Morley experiment. State the explanations for the negative results of the experiment. (CO3)
- (b) Write the postulates of special theory of relativity. Derive the expression for length contraction. (CO3)
- (c) (i) Derive Einstein mass energy relation. (CO3)
- (ii) Find the velocity of the particle if the total energy is thrice its rest mass energy.
4. (a) (i) Write a short note on Type I and Type II superconductor. (CO5, CO4)
- (ii) Discuss quantum well, quantum wire, and quantum dots on basis of quantum confinement, dimension, and delocalization.
- (b) Differentiate between top down and bottom up method used for synthesis of nanomaterials. Also discuss various methods used in each technique. (CO5, CO4)

(3)

- (c) Discuss the effect of temperature on the resistivity and external magnetic field of a superconducting material. Also show that perfect diamagnetism is a property exhibited by superconductor. (CO5, CO4)
5. (a) Derive Schrödinger time dependent wave equation. Calculate the velocity and kinetic energy of a neutron having de Broglie wavelength $1\text{\AA}$. ($h = 6.6 \times 10^{-34} \text{ J-s}$, $m = 1.67 \times 10^{-27} \text{ kg}$). (CO6)
- (b) Deduce the expression for normalized wave function for a particle moving inside a one dimensional box of infinite height. Also draw the graph for probability density of this particle at $n = 1, 2, 3$. (CO6)
- (c) Explain Heisenberg's Uncertainty principle. Calculate the smallest possible uncertainty in the position of an electron moving with a velocity of $3 \times 10^7 \text{ m/s}$. (rest mass of electron = $9 \times 10^{-31} \text{ kg}$). (CO6)

Roll No.

TEE-201

**B. TECH. (SECOND SEMESTER)
END SEMESTER EXAMINATION, June, 2023**

BASIC ELECTRICAL ENGINEERING

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Define following terms with suitable examples : (CO1, CO2)

(i) Bilateral and Unilateral elements

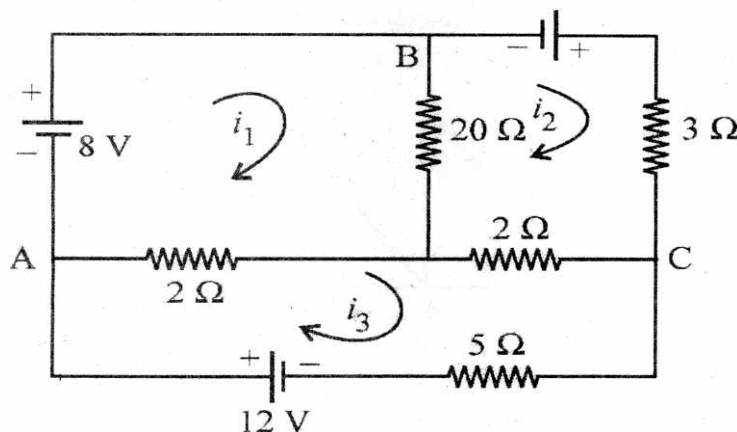
(ii) Supernode and Supermesh condition

(iii) Apparent and Active power in AC circuits

(iv) Ideal and practical current source

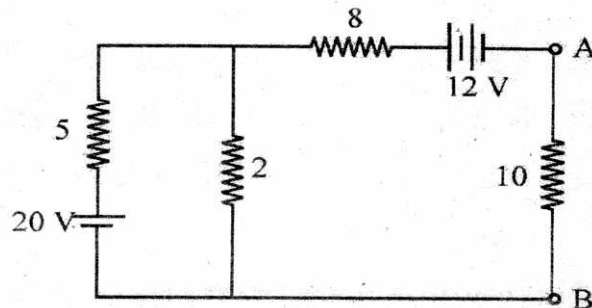
(v) Impedance triangle of a series R-LAC circuit

(b) In the network shown in Fig. 1, find the current across $5\ \Omega$ resistor using mesh analysis. (CO1, CO2)

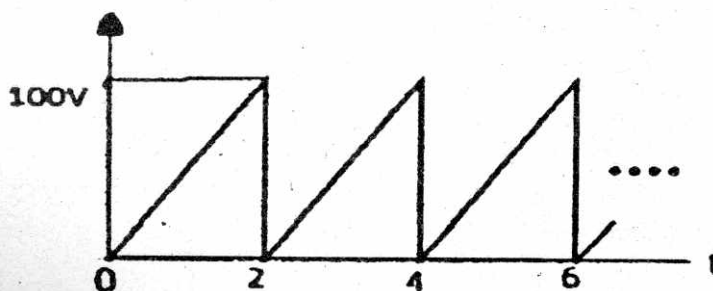


P. T. O.

- (c) State Thevenin's theorem. For the circuit shown in Fig. 2, calculate the current across $10\ \Omega$ resistor using Thevenin's theorem. (CO1, CO2)



2. (a) In a 2-terminal DC network, how the transformation of star-delta and delta-star takes place? With a proper derivation, find out the expression for star and delta parameters. (CO1, CO2, CO6)
- (b) Explain the working of single phase purely resistive circuit with proper phasor diagram. Also derive the expression for average power. (CO1, CO2, CO6)
- (c) A 6-pole lap connected generator has a useful flux/pole of $0.045\ \text{Wb}$. If the no load voltage at $400\ \text{r.p.m.}$ is $300\ \text{V}$, find the conductors on the armature periphery. (CO1, CO2, CO6)
3. (a) Calculate the average value, RMS value and Form factor of the sawtooth wave shown in Fig. 3. (CO2, CO3)



(3)

- (b) A 230 V, 50 Hz AC supply is applied to a coil of 0.06 H inductance and 2.5Ω resistance connected in series with a $6.8 \mu\text{F}$ capacitor. Calculate :

(CO2, CO3)

(i) Impedance

(ii) Current

(iii) Phase angle between current and voltage

(iv) Power factor

- (c) What do you understand by Electrical Wire ? Give classification of electrical cable. Explain any *one* in detail.

(CO2, CO3)

4. (a) What is the importance of grounding (earthing) in an electrical system ? Explain ? How many types of electrical earthing are possible ?

(CO4, CO5, CO6)

- (b) Give classification of electrical machines. Also explain the working principle of electrical motor.

(CO4, CO5, CO6)

- (c) Explain the term "Switchgear" in the electrical system. Give classification of switchgears. Explain moulded case circuit breakers (MCCB) in detail.

(CO4, CO5, CO6)

5. (a) Define voltage triangle, impedance triangle, and power triangle for single phase series R-C circuit.

(CO1, CO2, CO6)

- (b) Derive the condition of maximum power transfer in D.C. circuit. Also derive the expression for maximum efficiency in D.C. circuit at maximum power transfer condition.

(CO1, CO2, CO6)

- (c) Give classification of batteries ? Explain the characteristics of Lithium-ion batteries.

(CO1, CO2, CO6)

Roll No.

TCE-201

B. TECH. (SECOND SEMESTER)
END SEMESTER EXAMINATION, June, 2023

(All Branches)

BASIC CIVIL ENGINEERING

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

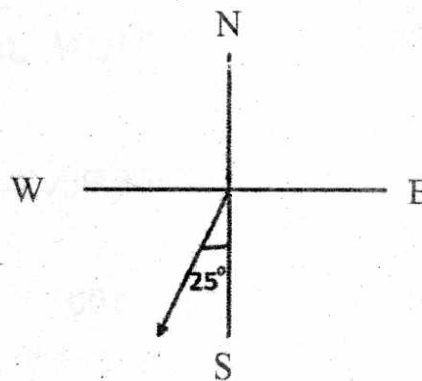
(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) List down *five* different types of civil engineering structures and their utilization to serve human establishment and society at large. (CO1)
- (b) Discuss the following branches of Civil Engineering in terms of their role and responsibilities : (CO1)
 - (i) Geotechnical Engineering
 - (ii) Environmental Engineering
 - (iii) Transportation Engineering
 - (iv) Water Resource Engineering
 - (v) Structural Engineering
- (c) What are the roles of civil engineer in the development of nation ? (CO1)

P. T. O.

2. (a) Distinguish between plans and maps. Define representative fraction. (CO2)
- (b) Discuss the classification of surveying based on instruments used. (CO2)
- (c) What is difference between Whole Circle Bearing and Quadrantal Bearing. Determine WCB and QB for the shown in fig. given below ? (CO2)



3. (a) Explain the term MORTAR. Classify it based on the type of binder used. (CO3)
- (b) An over dried brick was weighed as 3.5 kg. It was immersed in water for 24 hours and after that saturated surface dried brick weighed 3.85 kg. Calculate the water absorption of the brick. If the acceptable limit is 5%, whether the brick will be accepted for the construction purpose ? (CO3)
- (c) Discuss the following types of cement : (CO3)
- (i) Ordinary Portland Cement
 - (ii) Rapid Hardening Portland Cement
 - (iii) High Alumina Cement
 - (iv) Portland Pozzolana Cement
 - (v) Portland Slag Cement

(3)

4. (a) Briefly discuss the following types of buildings as per NBC : (CO4)
- (i) Educational Building
 - (ii) Business Building
 - (iii) Storage Building
 - (iv) Hazardous Building
 - (v) Assembly Building
- (b) Enlist the functions of foundation. Discuss the criteria for selection of suitable foundation. (CO4)
- (c) Discuss about the aspect of a building. Draw and mention the location of various parts of building based on sun diagram. (CO4)
5. (a) Sketch *five* regulatory signs, their name and meanings. Also, sketch *five* cautionary signs, their name and meanings. (CO5)
- (b) Enlist some applications of transportation engineering around you and how it is useful in your daily routine. (CO5)
- (c) Write down some water quality parameters along with the standard values assigned to them, if not what will be its impact on human. (CO5)

Roll No.

MAB-201

**B. TECH. (BIOTECHNOLOGY) (SECOND SEMESTER)
END SEMESTER EXAMINATION, June, 2023**

ENGINEERING MATHEMATICS

Time : Three Hours

Maximum Marks : 100

Note : (i) All questions are compulsory.

(ii) Answer any *two* sub-questions among (a), (b) and (c) in each main question.

(iii) Total marks in each main question are **twenty**.

(iv) Each sub-question carries 10 marks.

1. (a) Define matrix and find the inverse of $\begin{bmatrix} -4 & -3 & -3 \\ 1 & 0 & 1 \\ 4 & 4 & 3 \end{bmatrix}$, if invertible.

(CO1)

- (b) Find the eigen values and eigen vectors of the matrix :

(CO1)

$$\begin{bmatrix} -5 & 2 \\ 2 & -2 \end{bmatrix}$$

- (c) Define a general system of linear equations. Solve the following system of linear equations by matrix method :

(CO1)

$$x + y + z = 6$$

$$5x - y + 2z = 9$$

$$3x + 6y - 5z = 0$$

P. T. O.

2. (a) Define Ordinary Differentiation. Find the second derivative of following the function : (CO2)

$$y = e^{ax^2+bx+c} + (\log x)^5$$

- (b) A function is defined as follows. Using limits, examine its continuity : (CO2)

$$\begin{aligned} f(x) &= -x, & x < 0 \\ &= x, & 0 \leq x \leq 1 \\ &= 2 - x, & 1 < x \end{aligned}$$

- (c) Evaluate the following : (CO2)

(i) $\int x^2 e^{2x} dx$

(ii) $\int \frac{1}{x \log x} dx$

3. (a) What is differential equation ? Define order and degree of the differential equation. State the order degree of the following differential equation and solve : (CO3)

$$(x^2 y - 2xy^2) dx = (x^3 - 3x^2 y) dy$$

- (b) Define linear and non-linear differential equation with examples. State the standard form of linear differential equation and solve : (CO3)

$$\frac{dy}{dx} + \frac{4x}{(x^2 + 1)^1} y = \frac{1}{(x^2 + 1)^2}$$

- (c) Define linear differential equation of order 2 with standard form. State the solution procedure to solve the following differential equation : (CO3)

$$\frac{d^2 y}{dx^2} + 6 \frac{dy}{dx} + 9y = 2e^{-3x}$$